

Root Cause Analysis - Duplicate Registration Frontend Crash

One major issue discovered during testing was that the frontend application crashes when a user tries to register using an already existing email or username. Instead of displaying a proper validation message, the application throws an unhandled `TypeError` and shows a full error screen. The issue happens because the frontend Redux reducer assumes that `action.payload.errors` will always exist whenever an error occurs. However, in some cases the backend returns a null or differently structured response object. Since the reducer directly tries to access `.errors` on a null object, the application crashes immediately.

This creates a poor user experience because the user receives no meaningful feedback and cannot continue using the application normally. In a production environment, this could increase user frustration and reduce trust in the platform. The issue can be fixed by adding defensive null checks before accessing nested properties in the reducer. Proper frontend validation and standardized backend error responses should also be implemented.

Additionally, global error boundaries can be added to prevent the entire UI from crashing due to one failed request.